

Name

- For the ODE $2y'' - 6y' + 4y = 4\cos(2t)$

- Write down the complementary sol.

$y_c(t) =$

- Compute a particular sol.

$y_p(t) =$

- The Aug Mat for $y(0) = 1, y'(0) = 2$ is

- 1.** For the ODE $3y'' - 3y' - 18y = -3e^{4t}$

- 1.1.** Write down the complementary sol.

$y_c(t) =$

- 1.2.** Compute a particular sol.

$y_p(t) =$

- 1.3.** The Aug Mat for $y(0) = 2, y'(0) = 3$ is

- For the ODE $2y'' - 2y' - 12y = 2t^2 + 1$ with

- Write down the complementary sol.

$$y_c(t) =$$

- 1.4.** Give a y_p form and Aug Mat for the coefficients

$$y_p(t) =$$

$$\text{Aug Mat} =$$

- 2.** For the ODE $2y'' + 2y' - 24y = 2\cos(3t)$

- 2.1.** Write down the complementary sol.

$$y_c(t) =$$

- 2.2.** Give a y_p form and Aug Mat for the coefficients

$$y_p(t) =$$

$$\text{Aug Mat} =$$